

Predicting new research directions in materials science using large language models and concept graphs

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1 Concept extraction

2 Evaluating LLM-based Concept Extraction

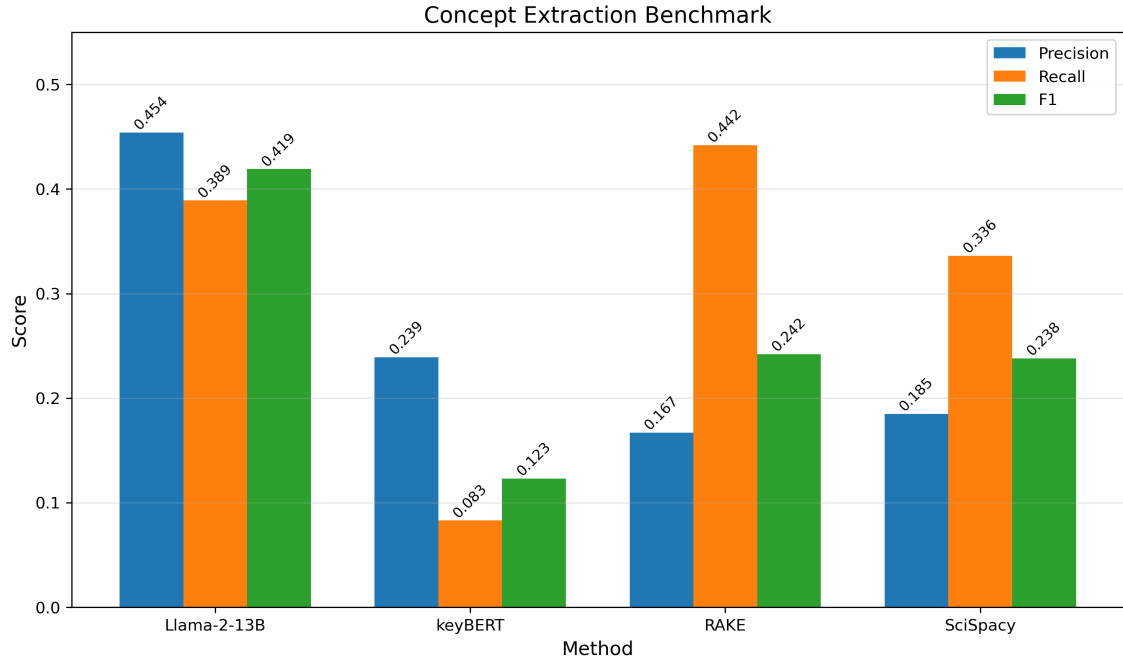
3 To construct a held-out test set, three annotators (n=3) manually annotated five abstracts that
4 were not part of the training data. The resulting inter-annotator agreement, measured as the mean
5 pairwise agreement, was 0.345 (pairwise scores: 0.215, 0.326, 0.493; all values rounded to three
6 decimal places).

7 Importantly, none of the human-annotated concepts were incorrect. Disagreement primarily
8 stemmed from varying emphasis on annotation breadth versus depth. As all extracted concepts
9 were valid, we defined the union of all annotations as the ground truth for evaluating our LLM-
10 based concept extraction approach against established methods, namely KeyBERT, RAKE, and
11 SciSpacy.

12 RAKE achieved slightly higher recall than the LLM-based approach, primarily due to its exten-
13 sive extraction of candidate terms. However, this also resulted in substantially lower precision, as
14 for every relevant concept, approximately five irrelevant ones were produced (precision $\approx 16.7\%$).
15 When balancing precision and recall, the LLM-based method outperformed all other approaches,
16 achieving a substantially higher precision at similar recall compared to RAKE.

17 Notably, concepts produced by the LLM that were not counted as true positives tended to
18 be semantically relevant but either overly broad or overly specific (e.g., ‘aerospace part thickness
19 measurement’), whereas competing methods frequently generated entirely irrelevant outputs such
20 as ‘6148 10 3 cycles’, ‘increase’, ‘interdisciplinary field involving principles’, or ‘317’. This highlights
21 the practical advantage of the LLM-based approach: it substantially reduces the need for manual
22 post-processing, as the extracted concepts are largely free of irrelevant entries.

23 To enable researchers to benchmark their own models, all labeled data can be found on our
24 GitHub repository.



Supplementary Fig. 1: Precision, Recall and F1-Score of our LLM-based approach, keyBERT, RAKE and SciSpacy; evaluated against a held-out test set of 5 annotated abstracts, including 138 ground-truth concepts in total.

Abstract

Removal of Metal Impurities from the Spent Hydroprocessing Catalyst and Acid Leaching Kinetics of Aluminum. The impurity metal of Al in spent hydroprocessing catalyst MoNi/Al₂O₃ was removed by HCl, and the metals of Mo and Ni were preliminarily enriched. The results show that the leaching efficiency of Al was 88.62%, and the leaching efficiency of Mo and Ni were 16.32% and 28.74%, respectively. The results were achieved under optimal leaching conditions: the particle size was 150 m, the concentration of HCl was 4 mol/L, the leaching temperature was, and the leaching time was 120 min. The kinetics of the leaching behavior of Al showed that the **acid leaching reaction of Al** was in accord with the equation $1-2/3/X-1-X^{2/3} = K_2 t$, $R^2 = 0.97734$, which was controlled by internal diffusion. X-ray diffraction analysis of the leaching residue revealed the existence of residual metals MoMoO₃, MoO₂ClO₄ and NiNiS. The separation of Al from Mo and Ni has been preliminarily realized, which is conducive to further efficient recovery of Mo and Ni.

LLM Concepts

acid leaching kinetic
aluminium acid leaching
 aluminium impurity
 aluminium removal
 hydroprocessing catalyst
 hydroprocessing catalyst impurity
 metal impurity
metal impurity removal
 spent hydroprocessing catalyst
 x ray diffraction analysis

Rake Concepts

acid leaching kinetics
 acid leaching reaction
 al showed
 efficient recovery
 equation xx
 impurity metal
 internal diffusion xray diffraction analysis
 leaching behavior
 leaching efficiency
 leaching efficiency
 leaching residue revealed
 leaching temperature
 leaching time
 metal impurities
 optimal leaching conditions
 particle size
 preliminarily enriched
 preliminarily realized
 residual metals momoo mooclo
 results show
 spent hydroprocessing catalyst
 spent hydroprocessing catalyst monialo

Supplementary Fig. 2: Concept Extraction for W3027426609

Abstract

Strengthening mechanisms in short **carbon fiber reinforced** Nb/Nb5Si3 composites with interfacial reaction. The understanding of strengthening mechanisms in composites is crucial to improve its mechanical properties. Strengthening mechanisms in Nb/Nb5Si3 composites reinforced by different contents of short carbon fiber C sf were investigated. Interface and tensile properties of the composites were studied. Interfacial nano-sized Nb3 phase forms by in situ interfacial reaction, the Nb3 phase embedded intimately in the C sf and matrix forms an anchor effect. The content of Nb3 phases in the composites increases because of the increase of C sf content. Ultimate tensile strength and yield strength of the composite are improved because of the increase of C sf content. When the content of C sf is 1.0 wt%, the elongation reaches the maximum, but the elongation decreases slightly when the content of C sf increases from 1.0 wt % to 2.0 wt %. The strengthening mechanisms in this composite were investigated according to grain refinement and load transfer. Load transfer is main strengthening mechanism and accounts for more than 95% of the improvement in yield strength. This indicates that the control of interfacial bonding state is crucial to improve the mechanical properties of the composites. This study can offer some valuable information to disclosing the strengthening behaviors in the C sf reinforced metal matrix composites.

LLM Concepts

anchor effect
carbon fiber reinforcement
 composite strengthening mechanism
 elongation
 grain refinement
 in situ interfacial reaction
 interface property
 interfacial reaction
 load transfer
 mechanical property
 nano sized phase
 short carbon fiber
 strengthening mechanism
 tensile property
 ultimate tensile strength
 yield strength

Rake Concepts

anchor effect
 c sf
 c sf content
 c sf content ultimate tensile strength
 c sf increases
 c sf reinforced metal matrix composites
 composites increases
 different contents
 elongation decreases slightly
 elongation reaches
 grain refinement
 interfacial bonding state
 interfacial reaction
 investigated according
 investigated interface
 load transfer load transfer
 main strengthening mechanism
 matrix forms
 mechanical properties
 mechanical properties strengthening mechanisms
 nb phase embedded intimately
 nb phases
 nbnsi composites reinforced
 short carbon fiber c sf
 short carbon fiber reinforced nbnsi composites
 situ interfacial reaction
 strengthening behaviors
 strengthening mechanisms
 studied interfacial nanosized nb phase forms
 tensile properties
 valuable information
 yield strength

Supplementary Fig. 3: Concept Extraction for W3192967258 (normalization of concepts through LLM extraction).

Abstract

Gas-Channel Design of Gas-Assisted Molded Parts. The structure design of parts in gas assisted injection molding GAIM was researched, while the parts with rectangular and semi-circular gas channels were studied by the methods of numerical simulation and physical simulation. At last, the degree of gas fingering was reduced numerical simulating experiment, CATIA was used to design simulated parts, and Moldflow was used to simulate the flow of plastic in the mold of GAIM. In physical simulating experiment, the dimension of gas-assisted injected plate was 160 mm 140 mm. Gas channels were decussated in the plate. Gas channels with different dimensions were designed to obtain the effects of gas channels on gas fingering. The results show that in the design of semi-circular gas channel there is optimized proportion range of radius to plate thickness and the gas fingering of parts with square gas channels is serious.

LLM Concepts

catia
flow of plastic
gas assisted injection molding
gas assisted molded parts
gas channel design
gas channel dimension
gas fingering
gas injected plate
moldflow
numerical simulation
physical simulation

Rake Concepts

design simulated parts
different dimensions
gas assisted injection molding
gas channels
gas fingering
gas assisted injected plate
gas assisted molded parts
gas channel design
mm mm gas channels
numerical simulating experiment catia
numerical simulation
optimized proportion range
physical simulating experiment
physical simulation
plate gas channels
plate thickness
results show
semicircular gas channel
semicircular gas channels
square gas channels
structure design

Supplementary Fig. 4: Concept Extraction for W2375662644.

Abstract

Microstructure and Wear Resistance of **Al₂O₃Coatings** on Functional Structure. To enhance the wear properties of function structure, Al₂O₃ -13%TiO₂ AT13 coatings were plasma sprayed on 45 steel functional structure using micro and nano powders. The microstructures and phase compositions of the coatings were investigated by scanning electron microscopy and X-ray diffraction, respectively. Results show that the nano powder coating consists of fully-melted region and partially-melted region. The fully-melted regions show a lamellar structure, while the partially-melted regions retain the powders structure. The phases of coatings are -Al₂O₃ and TiO₂. The wear test was carried out on a ML-10 friction and wear tester under dry sliding condition. It is found that the wear resistance of the micro powder coating is higher than that of nano powder coating. This is mainly ascribe to the breakage of the nano powder coating resulted from low agglomerated binding force.

LLM Concepts

agglomerated binding force
al₂o₃ coating
 coating microstructure
 dry sliding condition
 friction and wear tester
 functional structure
 lamellar structure
 partially melted region
 phase composition
 plasma spraying
 scanning electron microscopy
 wear property
 wear resistance
 wear tester
 x ray diffraction

Rake Concepts

dry sliding condition
 fullymelted region
 fullymelted regions show
 function structure alo tio
 functional structure
 lamellar structure
 low agglomerated binding force
 mainly ascribe
 micro powder coating
 ml friction
 nano powder coating
 nano powder coating consists
 nano powder coating resulted
 nano powders
 partiallymelted region
 partiallymelted regions retain
 phase compositions
 plasma sprayed
 powders structure
 scanning electron microscopy
 steel functional structure using micro
 wear properties
 wear resistance
 wear test
 wear tester
 xray diffraction respectively results show

Supplementary Fig. 5: Concept Extraction for W2521598419.

Abstract

Crystallization behaviour of AlSm amorphous alloys. Various Al100xSmx alloys 10 14 have been rapidly solidified by single-roller melt spinning with careful control of the atmosphere in the quenching device. The structural state and subsequent devitrification behaviour of the melt-spun ribbons are found to be particularly sensitive to the quenching conditions. Except for the thinnest ribbons, there are inhomogeneities both through the ribbon thickness and along the length. Both **fully and partially amorphous ribbons** have been obtained. The crystallization processes of the amorphous phases have been followed by X-ray diffraction, transmission electron microscopy and differential scanning calorimetry DSC. Al92Sm8 shows primary crystallization of Al, followed by the formation of metastable phases; Al90Sm10 transforms polymorphically to a metastable intermetallic; Al88Sm12 and Al86Sm14 display eutectic crystallization into Al and a metastable mixture of compounds. For Al90Sm10, the DSC traces involve several overlapping peaks. This may be the result of transformations occurring in distinct parts of the ribbon with different mechanisms. A kinetic analysis of the crystallization processes has been performed by means of isothermal and non-isothermal DSC experiments. A discussion of the kinetic parameters derived from Kissinger and Avrami analyses is provided.

LLM Concepts

amorphous alloy
avrami analysis
crystallization behavior
crystallization process
devitrification behavior
differential scanning calorimetry
dsc
eutectic crystallization
fully amorphous ribbon
inhomogeneity
isothermal dsc experiment
kinetic analysis
kissinger analysis
melt spinning
metastable phase
non isothermal dsc experiment
partially amorphous ribbon
polymorphic transformation
primary crystallization
quenching device
ribbon length
ribbon thickness
structural state
transmission electron microscopy
x ray diffraction

Rake Concepts

al followed
alsm amorphous alloys various alxsmx alloys
alsm display eutectic crystallization
amorphous phases
avrami analyses
careful control
crystallization behaviour
crystallization processes
different mechanisms
distinct parts
dsc traces involve several overlapping peaks
kinetic analysis
kinetic parameters derived
meltspun ribbons
metastable intermetallic alsm
metastable mixture
metastable phases alsm transforms polymorphically
nonisothermal dsc experiments
partially amorphous ribbons
particularly sensitive
quenching conditions except
quenching device
rapidly solidified
ribbon thickness
singleroller melt spinning
structural state
subsequent devitrification behaviour
thinnest ribbons
transformations occurring
xray diffraction transmission electron microscopy

Supplementary Fig. 6: Concept Extraction for W2047249259.

Abstract

Successive ionic layer adsorption and reaction SILAR trend for nanocrystalline mercury sulfide thin films growth. Mercury sulfide HgS nanocrystalline thin films have been grown onto amorphous glass substrate by successive ionic layer adsorption and reaction SILAR trend at room temperature 27C. The optimized preparative parameters including ion concentration, number of immersion cycles, and pH of the solution are used for fine nanocrystalline film growth. A further study has been made for the structural, surface morphological, optical and electrical properties of the films by using X-ray diffraction XRD, scanning electron microscopy SEM, transmission electron microscopy TEM, optical absorption and dc two point probe method. The as-deposited grown HgS nanocrystalline films exhibited cubic phase, with optical band gap E_g of 2.0eV and electrical resistivity of the order of 103cm. SEM and TEM images confirmed films of smooth surface morphology and nanocrystalline in nature with fine crystallites of 2030nm diameter, respectively.

LLM Concepts

amorphous glass substrate
cubic phase
dc two point probe method
electrical property
electrical resistivity
mercury sulfide
nanocrystalline mercury sulfide thin films
nanocrystalline mercury sulfide thin films growth
optical band gap
optical property
scanning electron microscopy
sem
successive ionic layer adsorption and reaction
successive ionic layer adsorption and reaction trend
transmission electron microscopy
x ray diffraction

Rake Concepts

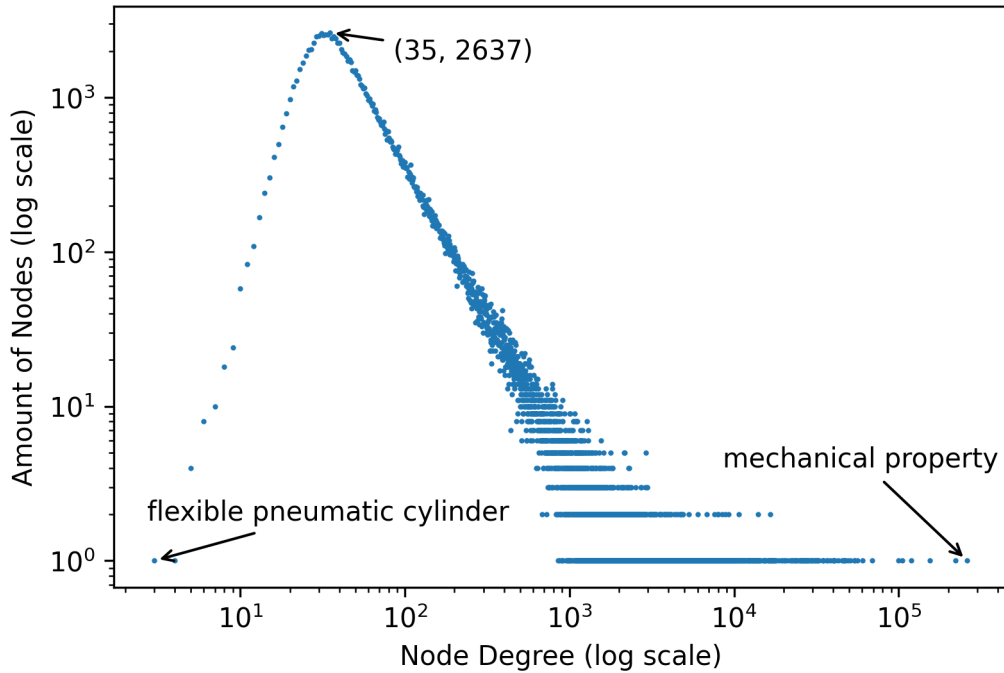
dc two point probe method
diameter respectively
electrical properties
electrical resistivity
fine crystallites
fine nanocrystalline film growth
grown onto amorphous glass substrate
immersion cycles
optical band gap eg
reaction silar trend
room temperature c
smooth surface morphology
structural surface morphological optical
successive ionic layer adsorption
tem images confirmed films

Supplementary Fig. 7: Concept Extraction for W1993498556 (correct interpretation of 'and'; abbreviation not extracted).

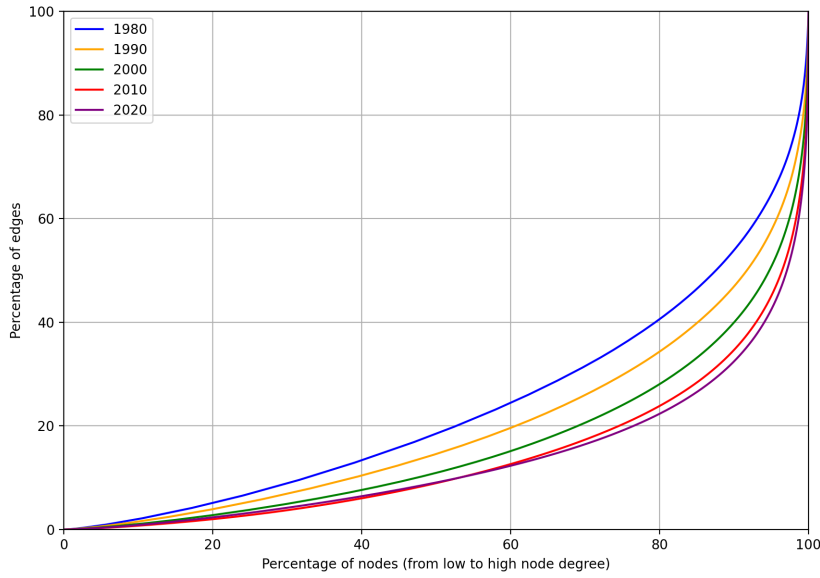
2 Concept graph properties

Supplementary Table 1: 25 most frequent concepts with their occurrences and degree in the concept graph.

Concept	Occurance	Degree
mechanical property	22931	261082
x ray diffraction	18021	221632
scanning electron microscopy	12185	154809
transmission electron microscopy	9462	118857
tensile strength	8840	105074
electron microscopy	8001	99775
SiC	6402	68771
grain size	5273	60612
aluminium alloy	5075	56272
grain boundary	4675	53671
yield strength	4202	54042
plastic deformation	4151	46884
optical property	4125	50866
TiO2	4085	49291
Al2O3	3988	47438
scanning electron microscope	3831	46055
tensile test	3814	46973
SiO2	3749	42820
ZnO	3706	45161
heat treatment	3538	40340
electrical property	3495	40945
tensile property	3362	40163
corrosion resistance	3219	38498
strain rate	3198	36590
magnetic property	3135	35066



Supplementary Fig. 8: Distribution of node degrees. The node degree of 35 is the most common, occurring 2637 times. The concept with the highest degree is 'mechanical property'. The concept with the lowest degree is 'flexible pneumatic cylinder'. Both axes are in log scale.



Supplementary Fig. 9: Concept Centralization over different years. The graph illustrates the cumulative percentage of edges connected to nodes when sorted by their degree. A steep curve indicates that a small percentage of nodes (highly central nodes) are connected to a large percentage of edges. The years 1980, 1990, 2000, 2010, and 2020 are compared to show how concept centralization increases over time in our Concept Graph.

Supplementary Table 2: 5 exemplary concepts and their respective 5 nearest neighbors

Concept	Neighbor	Distance
compound semiconductor	semiconductor compound	2.798
	compound semiconductor material	3.22
	semiconductor material	3.457
	semiconducting compound	3.584
	semiconductor crystal	3.658
pyrocarbon matrix	pyrocarbon layer	7.07
	resin carbon matrix	8.072
	pyrolytic carbon matrix	8.115
	matrix carbon	8.115
	carbon matrix	8.202
pure titanium sheet	pure titanium	5.801
	titanium sheet	5.82
	pure titanium plate	6.192
	commercially pure titanium sheet	6.594
	pure titanium surface	6.914
wear resistant coating	coating property	1.428
	corrosion resistant coating	1.465
	coating wear resistance	1.663
	coating protection	1.73
	coating characteristic	1.774
304l stainless steel	aisi 304l stainless steel	4.861
	316l stainless steel	5.294
	304l steel	5.323
	308l stainless steel	5.742
	type 304l stainless steel	5.834

27 3 Link prediction

28 Test set construction

29 To get a clear picture of our models’ performances, we do not perform oversampling for the positives
 30 during test set construction. This is necessary to analyze AUROC, precision and recall in a setting
 31 as close to real-world applications as possible. We approach test set construction by randomly
 32 sampling node pairs that are not connected until the end of 2019 and simply storing the label by
 33 looking three years ahead in the graph. The resulting 2,000,000 node pairs, therefore, mirror the
 34 real distribution of positives. Of these 307 (0.015%) positive examples, 87 links were formed in the
 35 year 2020, 123 in 2021, and 97 in 2022.

36 Evaluation against Science4Cast

37 We benchmark our Baseline model against the Science4Cast benchmark described in [14]. First,
 38 we train the model on data up until 2014 to predict links emerging in 2015, 2016 and 2017. Second,
 39 we evaluate on the node pairs contained in ‘SemanticGraph_delta_3_cutoff_0_minedge_1.pkl’.
 40 Surprisingly, our Baseline model is not only competitive but would have scored the second place
 41 in the Science4Cast competition with an AUROC of 0.9088. [Supplementary Table 3](#) presents an
 42 exhaustive overview of precision, recall, F1 score and the raw data at different thresholds.

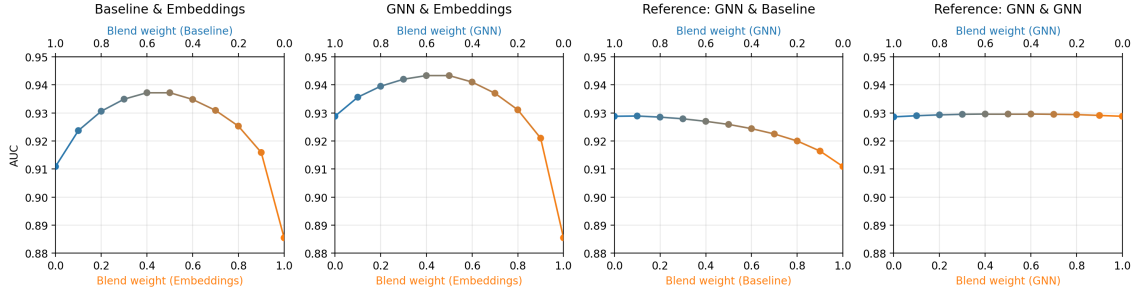
Supplementary Table 3: Performance metrics of our Baseline model at different thresholds evaluated against the Science4Cast benchmark on ‘SemanticGraph_delta_3_cutoff_0_minedge_1.pkl’ [14]

Threshold	AUC	Precision	Recall	F1-score	TN	FP	FN	TP
0.50	0.9088	0.0818	0.5170	0.1413	9663029	287423	23934	25614
0.55	0.9088	0.0952	0.4705	0.1583	9728873	221579	26238	23310
0.60	0.9088	0.1101	0.4266	0.1751	9779674	170778	28411	21137
0.65	0.9088	0.1267	0.3837	0.1905	9819397	131055	30538	19010
0.70	0.9088	0.1458	0.3410	0.2042	9851432	99020	32653	16895
0.75	0.9088	0.1693	0.2950	0.2152	9878748	71704	34930	14618
0.80	0.9088	0.2027	0.2497	0.2238	9901799	48653	37175	12373
0.85	0.9088	0.2474	0.1975	0.2196	9920681	29771	39763	9785
0.90	0.9088	0.3225	0.1332	0.1885	9936587	13865	42948	6600
0.95	0.9088	0.4881	0.0568	0.1017	9947501	2951	46734	2814

43 Analysis of Ensemble Performance Across Feature Domains

44 We investigated the impact of model blending on AUC performance, i.e. whether performance gains
 45 arise from combining distinct feature modalities or solely from the averaging effect of ensembling.
 46 While blends of models with complementary information (embeddings mixed with topological
 47 models, Panels 1 and 2 of [Supplementary Fig. 10](#)) achieve substantial performance boosts, blends
 48 within the same feature domain (Panel 3) or model type (Panel 4) conversely result in only marginal
 49 improvements. This demonstrates that the observed gains are driven by the synergy of non-
 50 redundant features rather than simple ensemble averaging.

51 Calibration of the models’ predictions

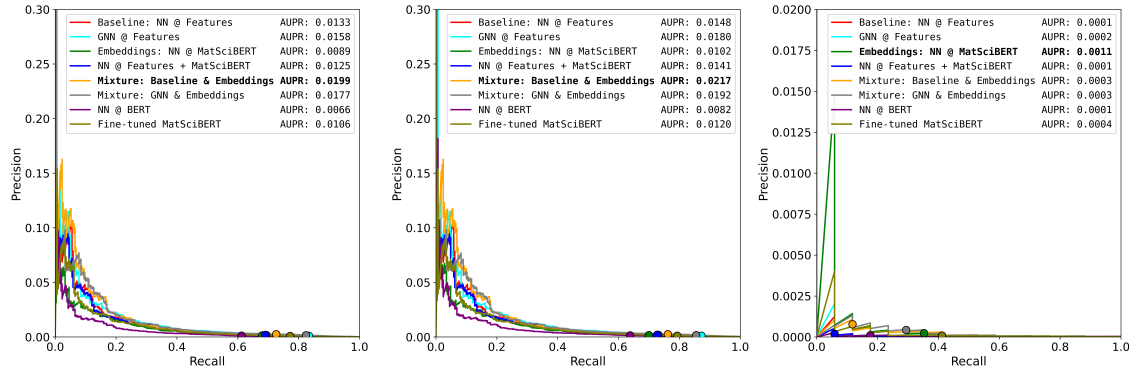


Supplementary Fig. 10: AUC performance for four hybrid models as a function of the blend weights for different combinations of models. We investigated whether performance gains arise from combining distinct feature modalities or solely from the averaging effect of ensembling.

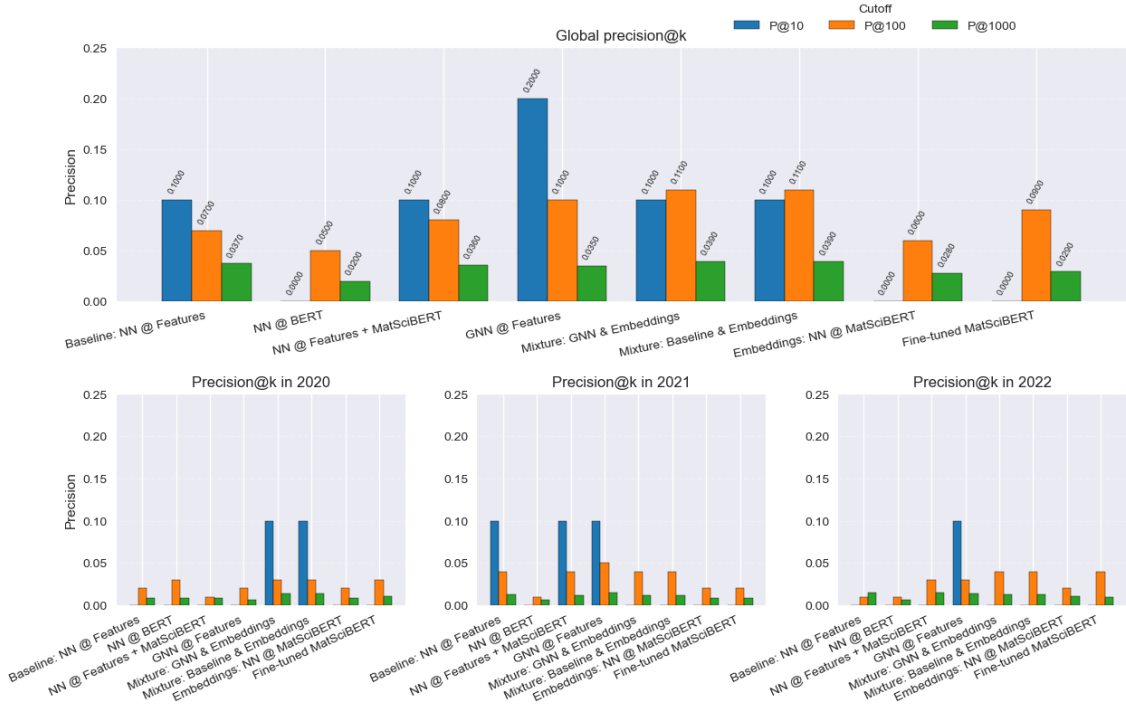
Supplementary Table 4: Calibration of our models' predictions measured with Brier-score.

Model	Brier Score
Baseline	0.0521
GNN	0.0953
Concept Embeddings (MatSciBERT)	0.0535
Concept Embeddings (BERT)	0.0596
Combination of features	0.05019
Mixture of MLP and Embeddings	0.0438
Combination of models	0.0438
Mixture of GNN and Embeddings	0.0438
Pure Text Baseline	0.1044

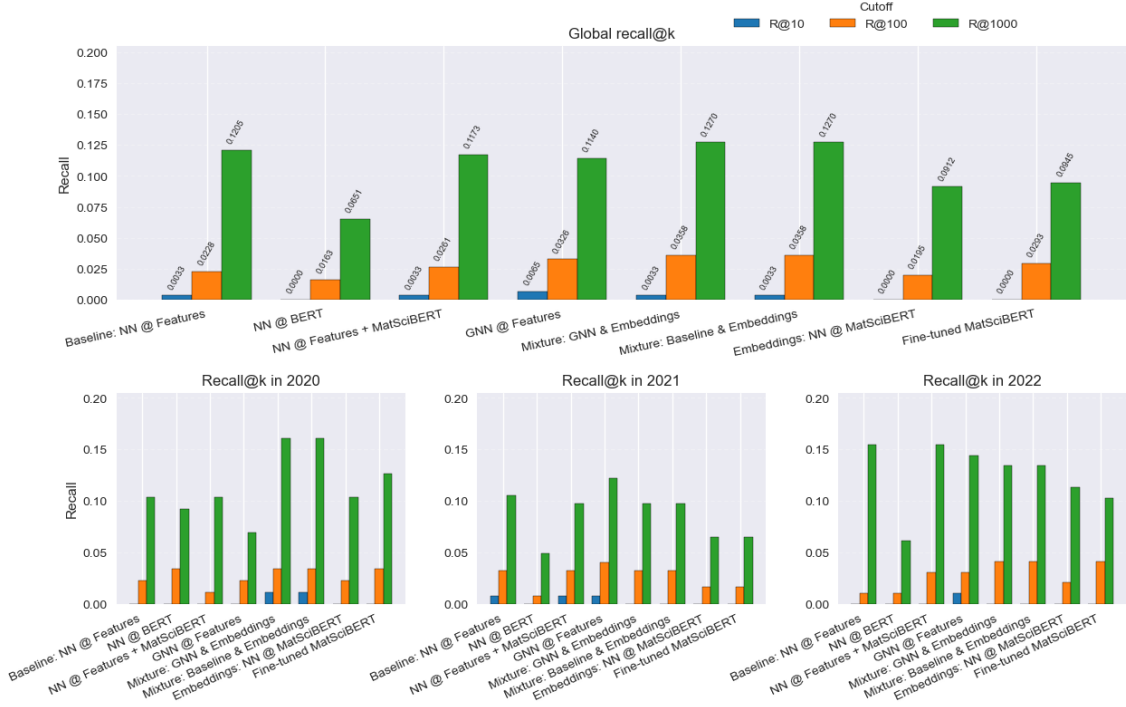
52 Precision and recall



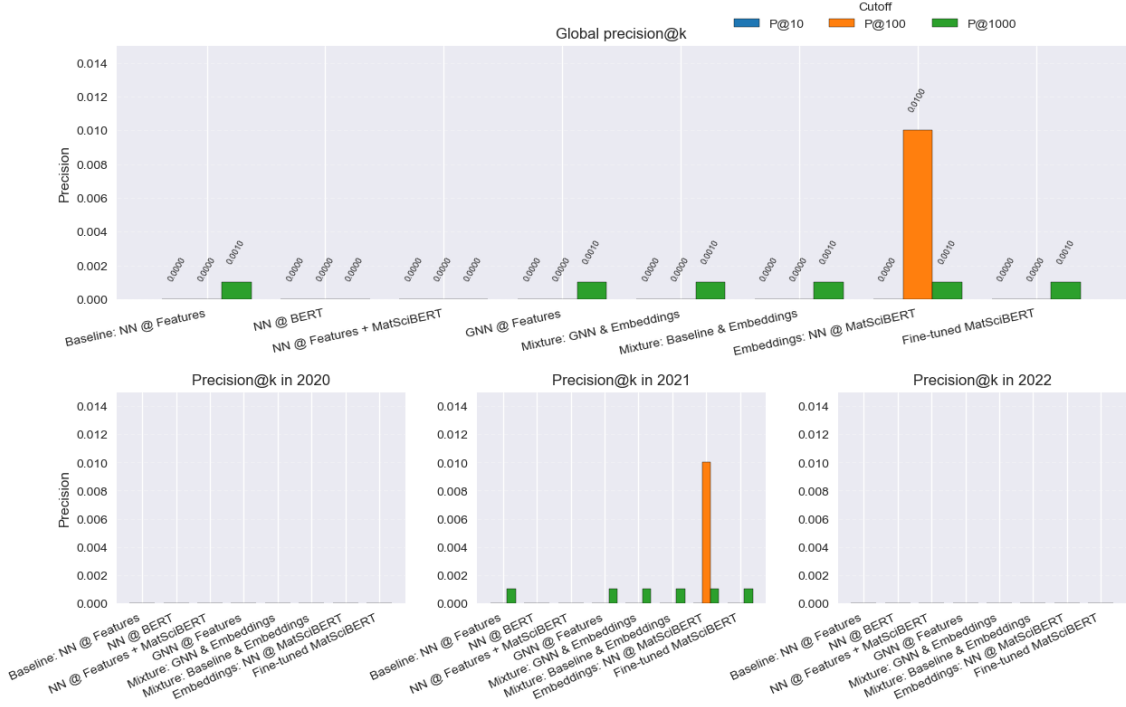
Supplementary Fig. 11: Precision-recall (PR) curves for our link prediction models on the test set ($T_{test} = [2020, 2022]$) with a zoom on the y-axis. Markers highlight the performances at a threshold of 0.5. The left panel shows the PR curve for all data points. The central and right panel display the respective PR curves for $d_{prev} = 2$ and $d_{prev} = 3$. Best result is in bold.



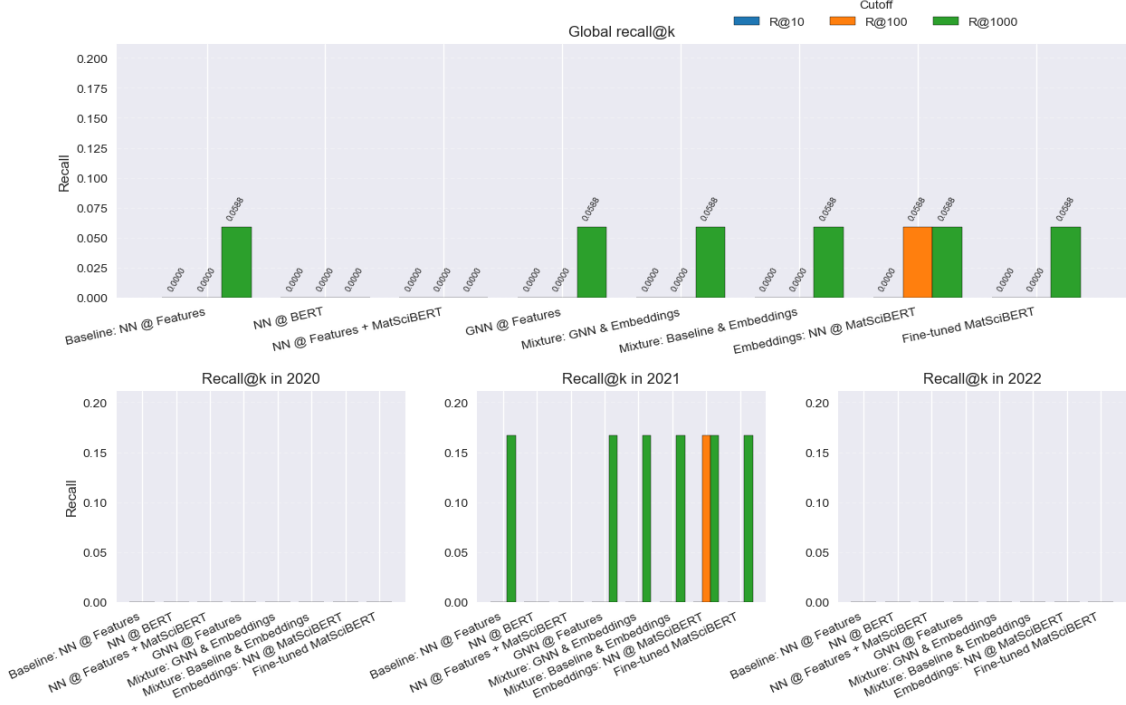
Supplementary Fig. 12: Precision@k for $k \in \{10, 100, 1000\}$ for our link prediction models for all data points, also broken down by year (2020, 2021, 2022).



Supplementary Fig. 13: Recall@k for $k \in \{10, 100, 1000\}$ for our link prediction models for all data points, also broken down by year (2020, 2021, 2022).

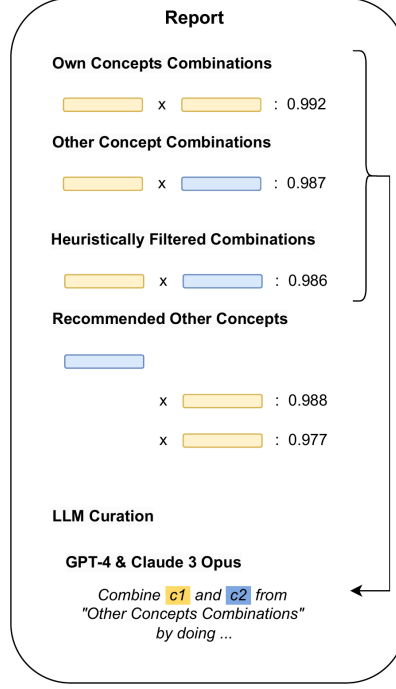


Supplementary Fig. 14: Precision@ k for $k \in \{10, 100, 1000\}$ for our link prediction models for $d_{\text{prev}} = 3$, also broken down by year (2020, 2021, 2022).



Supplementary Fig. 15: Recall@ k for $k \in \{10, 100, 1000\}$ for our link prediction models for $d_{\text{prev}} = 3$, also broken down by year (2020, 2021, 2022).

4 Report generation



Supplementary Fig. 16: Schematic representation of the report structure. 'Own' (C_{own}) and 'Other' (C_{other}) concepts are respectively represented in yellow and blue.

Own Concepts Combinations. We generate combinations by combining the concepts that describe a researcher C_{own} with themselves:

$$S_{\text{own} \times \text{own}} := \{(c_1, c_2) \in C_{\text{own}} \times C_{\text{own}} | (c_1, c_2) \notin E(G)\} \quad .$$

These combinations are then ranked, and the best 25 are included. The benefit of this section is that researchers are directly able to judge on the usefulness of these suggestions as both concepts are within their expertise. The size of C_{own} varies between researchers, we observe values roughly between 100 and 400.

Other Concept Combinations. We generate more diverse combinations by combining C_{own} with all remaining concepts:

$$S_{\text{own} \times \text{other}} := \{(c_1, c_2) \in C_{\text{own}} \times C_{\text{other}} | (c_1, c_2) \notin E(G)\}$$

where $C_{\text{other}} := C_{\text{known}} \setminus C_{\text{own}}$. These combinations are then ranked and the best 25 are included. Note that $|Combinations_{\text{own}, \text{other}}| \gg |Combinations_{\text{own}, \text{own}}|$ as the size of C_{known} is 137,000. The idea of this section is to offer more unbound suggestions than in 'Own Concepts Combinations'.

Filtered Other Concept Combinations. We employ two heuristics to filter through the large set of $S_{\text{own} \times \text{other}}$

$$S_{\text{own} \times \text{other}}^{\text{filtered}} := \{(c_{\text{own}}, c_{\text{other}}) \in Combinations_{\text{own}, \text{other}} | \deg(c_{\text{other}}) \leq 100 \ \& \ 3 \leq \text{dist}(c_{\text{own}}, c_{\text{other}} \leq 6)\} \quad .$$

To avoid unspecific suggestions, i.e. broad terms like 'materials science', we exclude concepts based on their *degree* in the concept graph G . To prevent obvious, e.g. synonyms, and too far-fetched concepts, we integrate semantic knowledge by using the euclidean distance in the map of the materials science (**Figure 1??**) that is a two dimensional projection of the obtained concept embeddings. Note that the values 3 and 6 were arbitrarily chosen after examining the map of materials science.

73 **Other Concepts x Many Own Concepts.** In this section, we suggest single concepts that
74 have highly scored connections with many own concepts C_{own} :

$$\begin{aligned} S_{(\text{many own}) \times \text{other}} &:= \{(c_{\text{other}}, C_{\text{own,matching}}) \in C_{\text{other}} \times \{C_{\text{own}} \times \dots \times C_{\text{own}}\} | C_{\text{own,matching}} \\ &:= \{c \in C_{\text{own}} | \text{score}(c_{\text{other}}, c) > p\} \end{aligned}$$

75 where p can be set freely as threshold. The resulting set is sorted descendingly according to the
76 length of $C_{\text{own,matching}}$ and the top 20 are included. The idea is to take the researcher’s profile into
77 account as a whole and to suggest concepts that can therefore be connected more easily to one’s
78 own research.

79 **LLM Curation** We are interested in the capabilities of LLMs to assist humans by pre-selecting
80 from a broader list of concept combinations. For this, we query two LLMs, GPT-4 and Claude
81 3 Opus (access May/June 2024), to select three items from a list of combinations and to write
82 an additional paragraph on why this combination is promising. Three combinations were selected
83 from $S_{\text{own} \times \text{own}} \cup S_{\text{own} \times \text{other}}$ together with three additional combinations from $S_{\text{own} \times \text{other}}^{\text{filtered}}$, resulting
84 in a total of six suggested combinations per LLM per report, and the two prompts are given in
85 Supplementary Fig. 17 and 18.

86 No paragraphs were created for combinations from section $S_{(\text{many own}) \times \text{other}}$ as we expected that
87 having the LLM generate a full paragraph combining multiple concepts would be harder to digest
88 and analyze within the limited interview time. Note that this report section does not contain new
89 suggestions but rather selects and elaborates on combinations already present in the report. Apart
90 from highlighting particular suggestions, the LLM approach helps to bridge the gap between two
91 concepts by providing additional information on how they could be combined. This is especially
92 helpful for $S_{\text{own} \times \text{other}}$ or $S_{\text{own} \times \text{other}}^{\text{filtered}}$ where the second concept might be unfamiliar to a researcher.

93 As our prompt to select 3 combinations of concepts from section 3 (Supplementary Fig. 18)
94 explicitly asked the LLM to choose “more exotic” combinations, we analyzed if this resulted in
95 a bias towards far-distance, cross-field pairs. For Sections 1 and 2, the mean node degree of the
96 selected pairs was slightly lower (9,712) than for the non-selected pairs (11718). While all of the
97 selected pairs were previously connected through only one additional node ($d_{\text{prev}} = 2$), 2 of the
98 non-selected pairs showed a larger previous node distance of $d_{\text{prev}} = 3$. For section 3, mean node
99 degrees of 1,582 and 4,694 were found for the selected and non-selected pairs, respectively. Here, 4
100 of the selected and 3 of the non-selected pairs showed a previous node distance of $d_{\text{prev}} = 3$. While
101 the lower node degrees as well as the slightly higher number of more distant concepts selected from
102 section 3 could result from the wording in the prompt, the small sample size prevents any definitive
103 conclusions.

Supplementary Table 5: LLM suggestion vs. "Is Interesting" (labeled as C) confusion matrix.

		LLM suggestion		
		Y	N	Σ
"Is Interesting"	Y	24	37	61
	N	29	24	205
	Σ	53	213	266

You are an advanced AI agent who has expertise in the realm of the materials sciences. Your goal is to assist researchers looking for promising new concept combinations that will --- once investigated --- advance mankind.

We prepared a report containing several concept combinations suggested by an ML model we trained on historical data. This is an overview of the attached PDF:

- 1.1 Possible new combinations of the concepts we identified for a particular scientist (which, to our knowledge, have not yet been combined)
- 1.2 Possible new combinations of the concepts we identified for a particular scientist with other existing concepts that, to our knowledge, the scientist has not yet worked on (at least not according to the abstract data of the last years)

Which 3 combinations (across both categories) do you think (including all material science knowledge you have) are the most promising? The order of the combinations doesn't indicate significance. Write a concise paragraph for each of those suggestions that explains why you chose this combination. Each suggested combination must be contained in the report. Don't invent new combinations that are not present.

Just give me the 3 combinations and the 3 paragraphs.

Supplementary Fig. 17: Prompt for the selection of 3 concept combinations from report sections 1 ($S_{\text{own} \times \text{own}}$) and 2 ($S_{\text{own} \times \text{other}}$).

You are an advanced AI agent who has expertise in the realm of the materials sciences. Your goal is to assist pairs of researchers looking for promising areas to combine their research which will - once investigated - advance mankind.

We prepared a report containing several concept combinations suggested by an ML model we trained on historical data. This is an overview of the attached PDF:

1. Possible new concept combinations that are more exotic.

For each combination, imagine two researchers proficient in one of the two fields/concepts that want to collaborate.

Which of these 3 of these more exotic combinations do you think (including all material science knowledge you have) are the most promising?

The order of the combinations doesn't indicate significance. Write a concise paragraph for each of those suggestions that explains why you chose this combination, and extend the paragraph by thinking about how the two scientists would collaborate and develop ideas and pathways to establish this cooperation.

Each suggested combination must be contained in the report. Don't invent new combinations that are not present.

Just give me the 3 combinations and the 3 paragraphs.

Supplementary Fig. 18: Prompt for the selection of 3 concept combinations from report section 3 ($S_{\text{own} \times \text{other}}^{\text{filtered}}$).

Nanoporous Structure and High Entropy Alloy

Nanoporous structures combined with high entropy alloys (HEAs) represent an intriguing area of study due to the unique properties HEAs offer, such as exceptional mechanical strength, corrosion resistance, and thermal stability. The introduction of a nanoporous architecture can potentially enhance these properties by introducing lightweight characteristics and increasing surface area, which could be particularly beneficial for applications in catalysis and energy storage. Combining these two concepts could lead to the development of new, optimized materials with tailored properties for specific industrial applications.

In Situ High Resolution Transmission Electron Microscopy and Mechanical Performance

The application of in situ high-resolution transmission electron microscopy (HRTEM) to study the mechanical performance of materials offers a promising research direction. This technique allows real-time imaging of material deformation, phase transformations, and dislocation dynamics at the atomic scale during mechanical testing. Understanding these microscopic events in great detail will enable the design of materials with superior mechanical properties tailored to specific applications, such clarification provides deeper insights into materials behavior under stress and can lead to the development of materials with enhanced durability and performance.

Epitaxial Strain and Ultraviolet Visible Spectroscopy

Studying the effects of epitaxial strain using ultraviolet-visible spectroscopy could open new pathways in the synthesis and characterization of thin films and nanostructures. Epitaxial strain has been known to significantly alter the electronic and optical properties of materials. By applying UV-Vis spectroscopy, researchers can gain valuable insights into the changes in band structure and electronic transitions caused by strain. This combination could be particularly valuable in the development of advanced optical devices, including lasers, LEDs, and photovoltaic cells, where precise control over electronic properties is crucial.

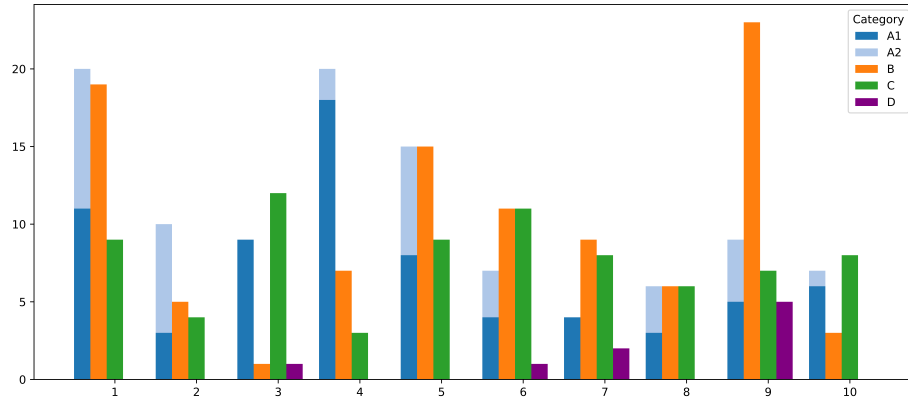
Supplementary Fig. 19: Example of an LLM output containing three concept combinations and the corresponding abstracts

5 Rated Concept Combinations

Selection of the interviewees

The selection of 10 interviewed researchers included only PIs, mostly professors at universities, from a diverse set of subdisciplines of materials science, some of which, but not all, had previous research experience with machine learning models.

Analysis of the rated concepts per researcher



Supplementary Fig. 20: Overview of the concept categories per interview. A1 and A2 are grouped together. Note that the results from researcher 3 were obtained by a commentary on the report, not by an interview.

Supplementary Table 6: Category counts per researcher with per-participant variance.

Researcher	A1	A2	B	C	D	Variance
1	11	9	19	9	0	36.64
2	3	7	5	4	0	5.36
3	9	0	1	12	1	24.24
4	18	2	7	3	0	41.20
5	8	7	15	9	0	22.96
6	4	3	11	11	1	17.60
7	4	0	9	8	2	11.84
8	3	3	6	6	0	5.04
9	5	4	23	7	5	51.36
10	6	1	3	8	0	9.04

111 **Examples of rated concept combinations**

Supplementary Table 7: Concept combinations ranked A1 and A2 during the interviews (25 examples per category)

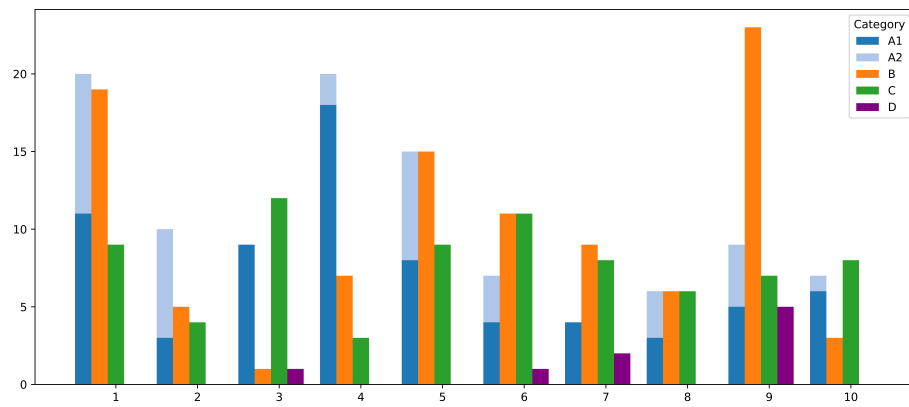
Category	Concept A	Concept B	Section ^[a]
A1	nanoparticle dispersion	quantum yield	3
	ferroelectric polarization	layered material	3
	in plane polarization	microstructure feature	3
	nanoparticle dispersion	post heat treatment	3
	microstructure characterization	molecular architecture	4
	energy dispersive x ray spectroscopy	domain structure	4
	ferroelectric polarization	thermal treatment	1
	grain texture	power conversion efficiency	1
	out of plane polarization	optoelectronic application	1
	polymer fullerene solar cell	kelvin probe force microscopy	1
	thermal treatment	crystal orientation	1
	computational material science	corrosion resistance	2
	ion extraction	density functional theory calculation	2
	solid electrolyte interphase layer	finite element analysis	2
	device architecture	CO2	2
	aging process	density functional theory	1
	solution concentration	machine learning	1
	acceptor material	crystal structure	1
	excitation intensity	thermal stability	1
	lattice distortion	optical analysis	3
	absorption spectroscopy	additive manufacturing technology	3
	artificial neural network	micro structural effect, crack initiation	4
	crystalline coating	laser powder bed fusion	2
	morphological change	composite material	2
	carbon enrichment	scanning electron microscope	2
A2	electrode fabrication	oxygen concentration	3
	optical property	high entropy alloy	2
	optoelectronic application	mechanical performance	2
	bulk heterojunction	thermal annealing	1
	circular dichroism	optoelectronic property	1
	electrode layer	quantum efficiency	1
	external quantum efficiency	thermal treatment	1
	fullerene derivative	optoelectronic property	1
	high reflectivity	perovskite solar cell	1
	pseudocapacitive material	composite material	2
	channel doping	x ray diffraction	1
	channel doping	x ray photoelectron spectroscopy	1
	chargedischarge test	thermal stability	1
	industrial application	surface pattern, cell membrane	4
	solid electrolyte interphase	structural property	1
	void filling	atom probe tomography	1
	resistance change	mechanical performance	2
	dislocation plasticity	gradient structure	3
	photon flux	high density	3
	edge defect	carbon nanotube	2
	femtosecond laser	density functional theory	1
	light irradiation	density functional theory	1
	multimodal imaging	photoluminescence spectroscopy	1
	solution synthesis	material design	1
	three dimensional printing	optical property	1

[a] Report sections: 1 ($S_{\text{own} \times \text{own}}$), 2 ($S_{\text{own} \times \text{other}}$), 3 ($S_{\text{own} \times \text{other}}^{\text{filtered}}$), 4 ($S_{(\text{many own}) \times \text{other}}$)

Supplementary Table 8: Concept combinations ranked A1 and A2 during the interviews (25 examples per category)

Category	Concept A	Concept B	Section ^[a]
B	photovoltaic application	microscopic analysis	3
	tailored property	deformed grain	3
	nanoparticle dispersion	energy absorption capacity	3
	material library	tension compression asymmetry	3
	optical property	mendelevium simulation	3
	crystal orientation	diffusion kinetic	3
	ferroelectric domain	interface bonding	3
	electrostatic repulsion	NS	3
	charge carrier transport	hydrophobic surface	3
	OH	crystal phase	4
	absorbance spectroscopy	tensile test	2
	polymer fullerene solar cell	laser powder bed fusion	2
	exciton diffusion length	x ray photoelectron spectroscopy	2
	cone beam	high entropy alloy	2
	cone beam	optoelectronic application	1
	hansen solubility parameter	optoelectronic application	1
	light element	optical property	1
	microfluidic chip	band gap	1
	rare earth oxide	electrostatic interaction	1
	artificial intelligence	BO	3
	ultimate tensile strength	gas evolution, material library, sub threshold swing	4
	spectroscopic technique	yield strength	2
	cycling performance	solar cell	1
	lithium ion battery application	high entropy alloy	1
	light conversion	mechanical property	2
C	charge carrier transport	hydrophobic surface	3
	conjugated polymer	elastic recovery	3
	electron backscatter diffraction	solar cell fabrication, thin film deposition	4
	machine learning	interpenetrating network, fluorescence quenching	4
	mechanical performance	polymer fullerene solar cell, ferroelectric property	4
	carbon based material	photocatalytic activity	2
	charge carrier transport	compressive strength	2
	crystal orientation	organic solar cell	1
	in plane polarization	organic solar cell	1
	mechanochemical reaction	tensile performance	3
	delafossite structure	dual phase	3
	dislocation accumulation	capacity fade, high entropy	4
	conventional ceramic	graphene oxide	2
	high entropy alloy	charge separation	4
	microstructure evolution	economical advantage	4
	semiconductor substrate	microstructure evolution	2
	mid infrared	molecular dynamic simulation	2
	exciton binding	hcp phase	3
	sodium iodide	microstructural defect	3
	metal organic framework	good ductility	4
	direct laser writing	youngs modulus	1
	graphene oxide	monte carlo simulation	4
	oxide transistor	cyclic voltammetry	2
	oxygen vacancy	crack tip plasticity	4
	stress induced phase transformation	hexagonal boron nitride	3
D	power consumption	nanoparticle formation	3
	quantum efficiency	additive manufacturing	2
	bauschinger effect	ultimate tensile strength	2
	moores law	high entropy alloy	1

[a] Report sections: 1 ($S_{\text{own} \times \text{own}}$), 2 ($S_{\text{own} \times \text{other}}$), 3 ($S_{\text{own} \times \text{other}}^{\text{filtered}}$), 4 ($S_{(\text{many own}) \times \text{other}}$)



Supplementary Fig. 21: Overview of the concept categories per interview. A1 and A2 are grouped together. Note that the results from researcher 3 were obtained by a commentary on the report, not by an interview.

6 Detailed discussion of interesting concept combinations

The selection of 10 researchers included only PIs, mostly professors at universities, from a diverse set of subdisciplines of materials science, some of which, but not all, had previous research experience with machine learning models.

Below is a more detailed discussion of the 5 selected examples of new combinations of concepts shown in the results section of the main text.

Suggestion: "Conventional ceramic" + "Graphene oxide"

During the concept prediction phase of the AI model trained on open-access scientific abstracts, an interesting connection between two research areas emerged. The AI identified "conventional ceramics" and "graphene oxide" as a potentially promising combination. While the integration of graphene oxide with ceramics has been explored in numerous studies, these approaches typically rely on different synthesis methods and subsequent mixing. In contrast, the method presented here is both unique and simple, yielding a distinctive material: a nanometer-thin multilayer graphene scaffold coated with an approximately 200 nm thick iron oxide layer (see **Extended Data Fig. 2??**). This AI-predicted research direction aligned with prior experimental work that had been conducted but never published, highlighting the model's potential to identify overlooked but valuable material synthesis strategies.

At first glance, the connection between conventional ceramics and graphene oxide is not obvious due to their fundamentally different structures and chemical behaviors. Conventional ceramics are typically ionic solids, composed of various cation-anion combinations (e.g., transition metals as cations and oxides, fluorides, or sulfides as anions). They are known for their rigidity, thermal and chemical stability, and widespread use in electronic and electrochemical applications. In contrast, graphene oxide consists of covalently bonded carbon and oxygen atoms, forming a highly flexible material that exfoliates and decomposes at elevated temperatures, producing carbon oxide species and exfoliated (multilayer)graphene.

The link between these material classes becomes evident when they are combined into a composite system. However, this integration is challenging due to their distinct synthesis methods and reactivity. In previous experiments aimed at developing a high-conductivity electrode material for conversion type Li-ion batteries, a successful combination of these materials was achieved. Graphene oxide was mixed with iron pentacarbonyl ($\text{Fe}(\text{CO})_5$), a liquid at ambient conditions, and subjected to autoclave treatment at temperatures above 250 °C. Under these conditions, $\text{Fe}(\text{CO})_5$ vaporized and decomposed into atomic Fe(0) species in the gas phase at around 200 °C. These iron atoms subsequently reacted with the oxygen groups in graphene oxide, forming iron oxide species (Fe_2O_3 , Fe_3O_4) while simultaneously reducing graphene oxide to graphene.

The resulting composite featured a highly conductive multilayer graphene scaffold coated with an approximately 200 nm thick layer of iron oxide. This material exhibited high electrochemical capacity and reversibility, making it a strong candidate for conversion-type electrode materials. The improved redox kinetics were attributed to the high electronic conductivity of graphene and the intimate contact with the thin iron oxide layer, enhancing the overall electrochemical performance.

Suggestion: "Tensile strain" + "Molecular architecture"

Managing tensile strain has been underestimated in the design of stable interfaces for thin film soft matter' solar cells. In inorganic crystalline semiconductors, strain / stress management has been understood as an important design criterion to prevent the formation of point defects, which are triggered by both mechanical and thermal stress. However, in organic and perovskite solar cells, the alternating combination of organic and inorganic interface materials are more and more being established as a design criterion due to the advantage of orthogonal solubility in multilayer processing. Tensile strain then arises at the interface of materials with differing thermal expansion coefficients during temperature fluctuations. Such fluctuations dominantly occur during processing, annealing, and post-crystallization. The mismatch in thermal expansion coefficients can cause macroscopic delamination or cracking at interfaces as well as microscopic formation of point defects such as halide voids.

Furthermore, the molecular flexibility of interface materials, particularly their torsional flexibility, significantly affects the mechanical properties of thin films, including the Young's modulus.

Molecules with higher torsional flexibility can accommodate strain more effectively, enhancing the mechanical resilience of the interface.

Recent research underscores that molecular tensile strain management might be regarded as a more general proxy for interface design. Brabec and Friederich reported in Science 2024 [31] torsional flexibility of HTL molecules, which was represented by the introduction of TPA units, as one of several design criteria for efficient perovskite solar cells [31]. We note that the suggestion "tensile strain + molecular architecture" was generated in December 2023. Properties such as film formation and homogenous film quality benefited from that as well. In another currently ongoing study by Brabec *et al.* (in submission), we found that embedding plasticizing molecules dramatically increases the mechanical and operational stability of thin film solar cells.

In conclusion, the effective management of tensile strain through the selection of interface materials with appropriate thermal expansion coefficients and molecular flexibility may have been underestimated for the design of long-time stable thin film organic and hybrid solar cells.

Suggestion: "Multiphase structure" + "Selective laser melting"

The term microstructure refers to the complete set of internal structural features of a material. This includes the crystallographic arrangement of its ideal, defect-free regions, as well as all types of defects that may be present. These defects range from point defects - such as impurity atoms, vacancies, and interstitials - to line defects like dislocations, planar defects such as grain boundaries, and three-dimensional defects including pores or precipitates. A well-characterized microstructure, from the atomic to the macroscopic scale, enables the establishment of correlations between structural imperfections and the mechanical or functional properties of materials. This understanding is essential for guiding the selection and application of materials in engineering contexts.

One important microstructural feature in complex metallic alloys and ceramic materials is the presence of phases - regions within the material that exhibit uniform physical and chemical properties, such as crystal structure and composition. Most technical materials are multiphase systems, meaning they contain multiple such regions. The formation of these multiphase structures typically results from processing steps that involve heating and cooling cycles, and often plastic deformation. These steps aim to achieve thermodynamically stable or kinetically stabilized microstructures. Controlling the multiphase structure is critical, as it allows materials to exhibit a desirable combination of properties, including enhanced strength, toughness, corrosion resistance, wear resistance, and optimized electrical or magnetic behavior.

Selective Laser Melting (SLM) is an advanced additive manufacturing (3D printing) technique in which a high-powered laser selectively melts and fuses fine layers of metal powder to fabricate solid, complex parts directly from a digital design. As the part is built layer by layer from metal powder, the process minimizes material waste while producing components with high detail and mechanical integrity.

Due to the rapid heating and cooling cycles inherent in SLM, the resulting microstructures are often far from equilibrium, frequently leading to the formation of multiphase materials. For example, parts produced via SLM may contain mixtures of phases with differing crystal structures or local compositions that would not typically form in slower, conventional manufacturing processes. These multiphase structures can significantly affect the material's final properties - potentially enhancing strength, hardness, or corrosion resistance, but also possibly introducing internal residual stresses that could compromise long-term performance.

Suggestion: "Stress induced phase transformation" + "Hexagonal boron nitride"

The link between stress-induced phase transformation and hexagonal boron nitride (h-BN) is best explained in the context of enhancing the fracture toughness of ceramics by phase transformations. For instance, in zirconia, a tetragonal-to-monoclinic phase transformation results in a volume expansion in the vicinity of a crack, which locally results in compressive stress both, deflecting and stopping crack growth. This concept is applied to other metastable phases in hard coatings [49]. A second concept of reaching an enormous tensile strength with simultaneously high fracture toughness, which both typically exclude each other, is using local anisotropy. This was shown in pearlite wires, where multiple micro-cracks parallel to the loading direction were formed and enhanced the fracture toughness substantially [50]. There are at least two arguments why thinking of stress-induced phase transformations and h-BN is reasonable. (i) Cubic boron nitride (c-BN), having a Wurtzite crystal structure, is used as a hard coating because of its enormous hardness.

In contrast, the hardness of h-BN is significantly lower, rendering it not a top candidate for hard coatings. On the contrary, h-BN offers a significant anisotropy in its mechanical properties and therefore might locally help to produce micro-cracks parallel to the loading direction, resulting in toughening. (ii) c-BN and h-BN have different densities, with c-BN exhibiting the higher one. I.e., upon compression, a certain propensity of forming c-BN exists. However, this phase transformation seems not to be trivial, and it remains unclear if the c-BN to h-BN phase transformation could be exploited from a materials engineering viewpoint. While first indications of improved mechanical properties by using a c-BN / h-BN composite do exist [51], the role of phase transformations to the best of our knowledge remains unknown for this system and could be an interesting topic for future evaluation.

Suggestion: "In-plane polarization" + "Organic solar cell"

A number of material properties were discovered in perovskite photovoltaics that have not been considered with respect to photovoltaics in the past, but which may benefit the generation and extraction of photogenerated charge carriers and hence the power conversion efficiency of the devices. Among these properties, a ferroelectric in-plane polarization of methylammonium lead iodide (MAPbI₃) was vividly discussed to help with the separation of electrons and holes as well as their conduction to the respective electrodes along ferroelectric domain walls [32,33]. So far, the ferroelectric properties are genuine to MAPbI₃ solar cells and some derivatives thereof [35]. The necessary precondition is a non-centrosymmetric unit cell of the crystal, which is why this concept has not played a role in any of the established solar technologies, such as silicon, so far [35,36]. Yet, it may well play a role in future novel perovskite-inspired materials [52].

The requirement of crystallinity and non-centrosymmetric unit cells seemingly does not link to organic solar cells, which are mostly amorphous. Yet, the great design flexibility of organic molecules can be used to introduce asymmetric polar functional groups, which are already in use today in piezoelectric polymers for organic touch sensors, e.g., polyvinylidene fluoride (PVDF). If, in addition, ways are found to orient these molecules, an organic analogy to in-plane polarization may be designed, which then also benefits the charge carrier separation in organic solar cells. The challenge here is that in-plane polarization is notoriously difficult to control via external electrical poling since the electrodes in organic solar cells only allow for the application of an out-of-plane electric field [37]. Hence, in-plane polarization probably must be part of the conformation of the film itself through post-processing or by molecular self-assembly. Covalent organic frameworks (COFs) may also mimic a crystal structure, which then can be modified for in-plane polarization.

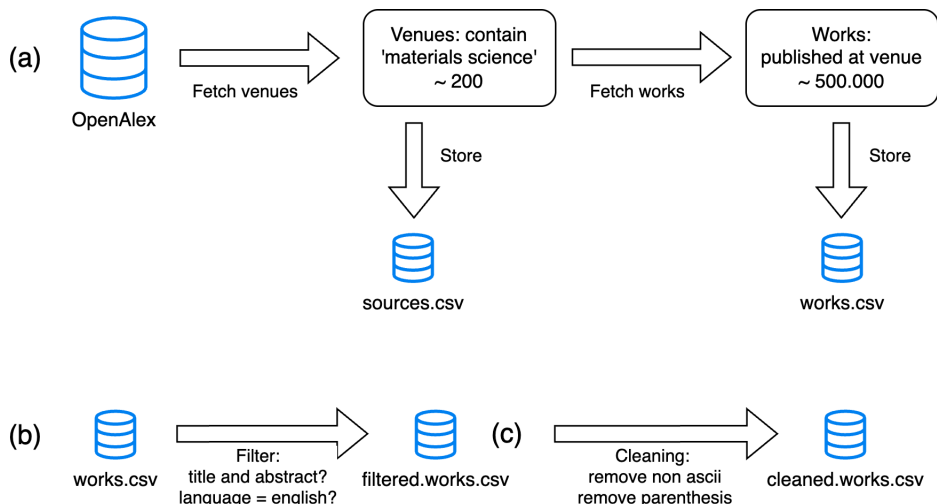
Furthermore, polarizable molecules typically have higher dielectric constants, which reduce the binding energy of photo-generated electron-hole pairs. While this binding energy still limits the performance of organic solar cells today, increasing the dielectric constant through polarization may help to enhance the open-circuit voltage of solar cells and to reduce monomolecular recombination. The idea of increasing the dielectric constant of organic molecules to reduce the binding energy of electron-hole pairs has been around for many years, yet without any major breakthroughs. The use of polarizing functional groups has not been thoroughly investigated so far.

As of today, both resulting concepts are rather speculative, but the machine learning suggestion points in an interesting research direction to be considered in future experiments.

7 Dataset

Filtering Papers fetched from venues related to materials sciences were first filtered, and only 259,913 out of 490,188 works had both a title and an abstract. Additional filtering removes very short (< 100 characters) and very long (> 4000 characters) abstracts. Furthermore, we check relatedness to the materials science domain by checking whether the tag 'Materials Science' is present. We rely on tags from OpenAlex's [tagger](#). We exclude works that contain Latex code and those whose primary language is not English.

Cleaning We exclude non-ASCII characters, remove tabs, multiple whitespaces and newlines. We use heuristic approaches to replace latex super or subscript code. We replace number ranges, parenthesis and brackets.



Supplementary Fig. 22: Overview of the dataset generation: (a) fetch publications from [OpenAlex](#) venues related to materials sciences, (b) filtering based on title and abstract (c) removal of selected characters.

Chemical Formulas Predominantly found in the materials science domain are material formulas. We employ a custom parser to recursively identify materials following these five rules:

1. The subsequent word is another element
2. The subsequent word is an integer together with a + or -
3. The subsequent word is an integer, a + or a -
4. The subsequent word is a decimal
5. The subsequent word is a composition of elements of the periodic table

The following examples showcase how those rules translate into real parsing: 'Na Cl' → 'NaCl', 'Na 2' → 'Na2', 'Mn 4+' → 'Mn4+', 'Fe3 O' → 'Fe3O', 'Fe3O 4' → 'Fe3O4', 'Mn 0.75' → 'Mn0.75', 'Zn3 PO' → 'Zn3PO', and 'H2 SO4' → 'H2SO4'.

8 LLM Fine-tuning

Hugging Face’s Python library was used to fine-tune the LLaMa-2-13B model [41,53]. We prepare our annotated data by constructing a single string per data point as follows `<s> + abstract + "#KEYWORDS#" + concepts + </s>` where `concepts` corresponds to the string version of a Python list, e.g. `['brittle epoxy', 'butadiene nitrile liquid rubber', ...]`. `<s>` and `</s>` represent the beginning of sequence (BOS) and, respectively, end of sequence (EOS) token for the LLaMa models. `#KEYWORDS#` serves as an own custom delimiter to signal the model when concept extraction should start. For the LLaMa tokenizer, we set `pad_token_id = 0`, `bos_token_id = 1`, and `eos_token_id = 2`. Being a CausalLM, we set the labels equal to the input token ids. Our `LoraConfig` is `r = 16` (Lora attention dimension), `lora_alpha = 32`, `target_modules=["q_proj", "v_proj"]` (modules where the adapter is applied), and `lora_dropout = 0.01` (dropout probability) [44][dettmers2023qlora]. Training took around an hour on a single A100 GPU.

9 Hyperparameters of the link prediction models

The GNN link prediction model uses a two-layer GraphSAGE encoder with a hidden dimension of 256 and produces 128-dimensional node embeddings. These embeddings are then fed into an MLP decoder that also has a hidden dimension of 256 and operates on 128-dimensional inputs. Neighborhood sampling is performed with fanout $(k_1, k_2) = (20, 15)$ for the first and second hop, respectively. The model is trained with a learning rate of $1 \cdot 10^{-5}$; additionally, a ReLU activation function, dropout with $p_{\text{dropout}} = 0.1$ is used.

Supplementary Table 9: Optimized hyperparameters of link prediction models.^[a]

Model	Layers	Neurons per layer	Learning rate
Baseline	7	[20, 300, 180, 108, 64, 10, 1]	$5 \cdot 10^{-4}$
Concept embeddings	5	[1536, 1024, 819, 10, 1]	$1 \cdot 10^{-3}$
Combination of features	7	[1556, 1556, 933, 559, 335, 10, 1]	$1 \cdot 10^{-3}$

[a] A ReLU activation function, a dropout probability of $p_{\text{dropout}} = 0.1$ and 30% of positive samples per batch were used for all models.

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